Acropolis Institute of Technology and Research, Indore
Department of Information Technology

AI for Career Development

GRIMOIRE

Learn. Interview. Get Hired.

Team BotS
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AI मंथन 2.0

GRIMOIRE: Autonomous AI Career Platform

Proposed Solution

- AI career coach from skill gap to job offer
- Diagnoses gaps with a smart test
- Builds a week-by-week roadmap
- Voice mock interviews with live code review
- Webcam feedback on posture and eye contact
- Tamper-proof badges and a public portfolio

How It Addresses the Problem

- Problem: students finish courses but freeze in interviews
- Human mock interviews cost ₹3,000+ per hour
- GRIMOIRE practises the real interview, voice plus code
- Roadmap adapts when scores drop
- Verified proof of work replaces keyword resumes

Innovation and Uniqueness

- One platform: learn, practise, get verified
- Multimodal: voice, vision and code together
- Ed25519-signed credentials employers can trust
- Link-checked learning resources
- Working prototype already built

Students don't fail from lack of knowledge. They fail from lack of practice under pressure.

BotS

TECHNICAL APPROACH

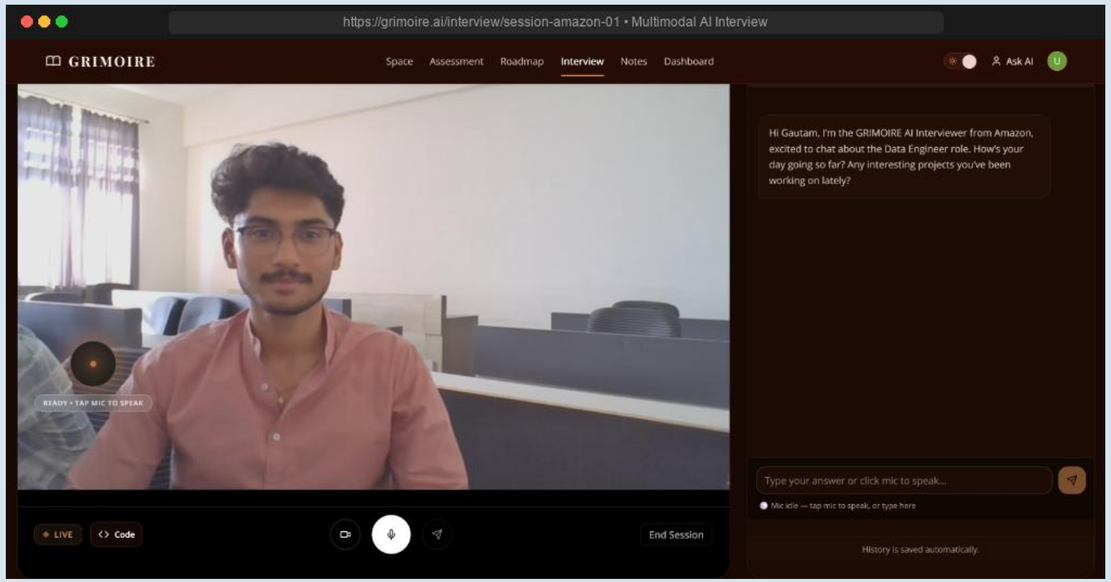
Technologies Used

Frontend	Backend	AI / LLM	Vision & Speech	Data
Next.js, React, TypeScript, Tailwind	Python, FastAPI, WebSockets	Groq (Llama 3), Gemini, Ollama	MediaPipe, OpenCV, Web Speech API	ChromaDB, Redis, Firestore, PostgreSQL

Methodology: Student Journey

1. **Diagnostic Assessment:** Skill-gap test & personalized baseline score
2. **Adaptive Learning Plan:** AI-curated roadmap with live resource verification
3. **Contextual Doubt Solver:** Multi-modal AI assistant for voice & code queries
4. **Multimodal Mock Interview:** Real-time voice STT/TTS, vision posture & coding simulator
5. **Verifiable Credential:** Ed25519 tamper-proof digital cryptographic badge

Working Prototype: Step 4 (AI Mock Interview Room)



FEASIBILITY AND VIABILITY

Feasibility	Challenges and Risks	Strategies to Overcome
• Full-stack prototype already working • Built on open-source tools and free-tier AI APIs • Estimated cost under ₹5 per session • Scales through campus training cells • Modular agents: each can be upgraded alone • Plan: pilot in colleges, then scale	AI API limits and cost	Multi-model failover: Groq, Gemini, local Ollama
	Webcam privacy	Consent first. Only posture and eye-contact scores are used
	Wrong or dead study links	Deterministic search plus link checks
	Weak internet in Tier-2/3 colleges	Offline fallback and browser-based speech

IMPACT AND BENEFITS

Target Audience

- Engineering and STEM students
- Tier-2 and Tier-3 colleges
- Training and placement cells
- Recruiters who want proof of skills

Social

- Mentoring for students who cannot afford it
- More confidence before real interviews

Economic

- About ₹5 per session vs ₹3,000+ per hour
- Faster, better-matched hiring

Institutional

- Standard placement training without outside agencies
- Progress data for colleges

Environmental

- Fully online coaching, no travel
- Paperless verified credentials

Estimated ₹5 per session vs ₹3,000+ per hour for a human mock interview

RESEARCH AND REFERENCES

Concepts and Research

Retrieval-Augmented Generation (Lewis et al., 2020)
arxiv.org/abs/2005.11401

Ed25519 signatures: RFC 8032
rfc-editor.org/rfc/rfc8032

Bloom's Taxonomy for adaptive difficulty
Anderson & Krathwohl, 2001

GRIMOIRE prototype
Working build, demo on request

Tools and Documentation

Google MediaPipe
ai.google.dev/edge/mediapipe

ChromaDB vector database
docs.trychroma.com

Groq inference API
console.groq.com/docs

FastAPI and Next.js
fastapi.tiangolo.com, nextjs.org/docs

Web Speech API (MDN)
developer.mozilla.org

Thank You

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